

Laguna State Polytechnic University – Los Baños Campus
College of Computer Studies

ITEP 310: Social and Professional Issues in Information Technology

Assessing the Effectiveness of Blockchain Technology in Improving Transparency in Government Administrative Processes In the Philippines

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Introduction/Background of the Case

Transparency in government remains a long standing issue in the Philippines. Many offices still rely on manual documentation, fragmented databases, and outdated record keeping systems, which make fund allocation, procurement, and project monitoring unclear to the public. These gaps limit citizens' ability to verify how government resources are used and weaken trust in public institutions. The lack of real-time visibility and reliable documentation creates opportunities for corruption and misinformation (Transparency International, 2023).

This issue directly affects multiple stakeholders. Citizens are left uncertain about government actions, public service agencies face delays and inconsistency in reporting, and government workers struggle with inefficient workflows and vulnerability to accusations due to incomplete or inaccessible records (Asian Development Bank, 2021). For IT professionals specializing in system development, digital services, and information management, these challenges highlight the urgent need for technological solutions that ensure traceability, security, and openness.

Studying this topic is essential from an IT and professional perspective because modern digital tools, especially blockchain, offer new ways to improve transparency, reduce manipulation of records, and support clean governance. Blockchain aligns with SDG 9: Industry, Innovation, and Infrastructure, which emphasizes stronger digital systems and supports the PDP 2023–2028 agenda on digital transformation, anti-corruption, and citizen-centered services.

Statement of the Ethical or Professional Issue

The central ethical and professional issue in this case is the persistent lack of transparency in government administrative processes in the Philippines. This deficit can lead to mismanagement of public resources, inefficiency, and reduced public trust. According to the Corruption Perceptions Index (CPI) 2023, the Philippines ranks 115th globally, highlighting ongoing transparency and accountability challenges (Transparency International, 2023).

The ethical dilemma arises from the conflict between the government's need to control internal administrative data and citizens' right to access accurate, timely, and verifiable information regarding government decisions, budgets, and project outcomes. Without transparency, auditing decisions becomes difficult, corruption opportunities increase, and public accountability weakens (World Bank, 2020).

Blockchain technology is proposed as a solution to enhance transparency and accountability by creating immutable, traceable, and publicly verifiable records. Studies indicate that blockchain-based governance models can improve transparency, reduce corruption risk, and support accountability in public administration (Berryhill et al., 2018; Samson et al., 2024; Ali et al., 2025). Its implementation, however, raises professional and ethical challenges, including equitable access, safeguarding sensitive information, and maintaining system integrity. IT professionals must design and implement digital systems that balance ethical standards, technical feasibility, security, and public interest, ensuring citizens' trust and responsible governance (World Bank, 2020; Transparency International, 2023).

Analysis and Discussion

The Philippine government's administrative processes—procurement, budgeting, and public records management—remain fragmented and vulnerable to manipulation. This lack of transparency affects multiple stakeholders: citizens rely on accurate information, government workers risk accountability issues due to incomplete documentation, and institutions struggle with oversight.

From a utilitarian perspective, adopting blockchain technology is ethically justified because it maximizes public good by providing tamper-proof records, reducing corruption risks, and enhancing trust. Verifiable and traceable information benefits both government agencies and the public by prioritizing accountability, data integrity, and digital efficiency (World Economic Forum, 2023).

This issue carries significant social and professional impacts. Citizens lose trust, government workers face accountability challenges, and institutions experience inefficiency and weak

oversight. In the long term, such gaps may perpetuate corruption risks and erode public confidence in government systems. Moreover, current practices conflict with policies like the Freedom of Information (FOI) Order and the Anti-Red Tape Act, which mandate accessible and reliable public records. Blockchain can support these mandates by providing secure, traceable, and verifiable information that strengthens accountability.

4.1 Stakeholders Analysis

The push for greater transparency affects a broad spectrum of stakeholders—citizens, public officials, oversight institutions, and IT professionals.

- **Citizens:** Limited access to complete, timely, and verifiable information undermines meaningful oversight and erodes trust in institutions (Transparency International, 2023). Blockchain can empower citizens to monitor government actions through immutable, publicly verifiable records of procurement, fund disbursement, and project completion.
- **Public Officials and Government Workers:** Fragmented records and manual processes increase the likelihood of errors and expose officials to unjust accusations (OECD, 2020). Blockchain provides transparent transaction logs with timestamps, streamlining workflows, protecting officials from false allegations, and facilitating audits.
- **Oversight Agencies:** Traditional portals often provide only partial information. Blockchain ensures end-to-end visibility of procurement life cycles and fund flows in a tamper-resistant ledger (PhilGEPS, 2025).
- **IT Professionals:** Their role is crucial in designing a secure, user-friendly system that balances technical feasibility, privacy, and equitable access, while maintaining ethical and social responsibility.

4.2 Ethical Framework Application

From a utilitarian perspective, adopting blockchain for government budgeting and procurement offers compelling benefits. It reduces opportunities for corruption, improves public trust, and enhances the efficiency of public service delivery, producing greater good for more people and outweighing implementation costs (World Economic Forum, 2023). Rights-based and deontological ethics further support this shift: the government has a moral duty to ensure transparency and accountability, while citizens have a right to accurate, timely information about the use of public resources (Transparency International, 2023). A blockchain-based system fulfills this duty by enabling immutable, publicly verifiable records, reinforcing trust in governance and safeguarding civic rights.

At a professional ethics level, IT and governance professionals must act with integrity, fairness, and responsibility. This entails designing systems that protect sensitive information, provide equitable access, and do not exclude or marginalize disadvantaged groups. Implementing blockchain in public administration demonstrates a commitment to these principles by promoting fairness through transparency, safeguarding data integrity, and advancing social justice through more open and accountable government operations.

4.3 Social and Professional Implications

Implementing a blockchain-based governance platform carries significant social and professional implications. Socially, it can help rebuild public trust in government by enabling citizens to monitor public funds, procurement processes, and project outcomes in real-time. This transparency fosters civic engagement and empowers communities to participate in governance beyond periodic elections, improved resource allocation, and more responsive public services, strengthening democratic governance.

Professionally, the shift encourages modernization of public-sector IT infrastructure and practices. IT professionals are challenged to innovate responsibly, upgrading skills and embracing ethical design principles. Public servants benefit from clearer workflows, fewer manual errors, and protection against false allegations. Oversight agencies gain more reliable

data, simplifying audits and compliance checks. However, the transition is not risk-free: equity concerns may arise if some citizens lack digital access or literacy, privacy issues may emerge if sensitive data is mishandled and over-reliance on technology could introduce vulnerabilities if systems are poorly maintained or attacked. These risks must be carefully mitigated through inclusive design, access support, robust cybersecurity, and fallback procedures for underserved populations.

4.4 Legal and Policy Context

Globally, governments have begun to adopt blockchain and distributed-ledger technologies to improve transparency and trust in public administration. For example, countries like Estonia and Georgia have successfully implemented blockchain in land registries, identity management, and other public records – improving security, reducing fraud, and increasing public confidence (UNECE, 2024; Coinomist, 2023). In the public procurement domain, initiatives documented by the World Economic Forum have demonstrated how blockchain-based bidding and contract systems can reduce corruption, improve auditability, and automate transparency vendors selection and fund disbursement processes (WEF, 2020; Samson et al., 2025).

In the Philippine context, existing transparency frameworks such as the Freedom of Information Order and the Anti-Red Tape Act reflect legal commitments to open governance and streamlined public service. Yet current systems like procurement portals often cover only partial aspects (e.g., bidding announcements and contract awards) without providing full visibility into fund disbursement, project execution, or real-time budget flows (PhilGEPS, 2025). By integrating blockchain, the proposed system addresses these gaps, aligning with legal mandates while adopting proven international practices that ensure data integrity, auditability, and public trust.

Proposed Solution and Recommendation

A feasible and ethical solution to the transparency issue is the development and implementation of a **blockchain-based government information system** designed to securely record and monitor administrative transactions such as budgeting and procurement. By storing data in immutable, time-stamped ledgers, the system minimizes opportunities for manipulation and

ensures that records remain verifiable and consistent across agencies. Leveraging the competencies of BSIT students in web and mobile application development, this system can be deployed through a structured digital platform that provides authorized users and citizens with reliable, blockchain-validated information.

This recommendation aligns with **SDG 9**, which promotes innovation and resilient digital infrastructure, and supports the **PDP 2023-2028** agenda on transparency and digital governance. Its feasibility lies in a phased implementation approach involving the digitization of existing workflows, establishment of strict access controls, and capacity-building for government personnel. Overall, this solution strengthens accountability, improves information integrity, and provides a long-term, sustainable strategy for enhancing transparency in government administrative processes.

Conclusion

The persistent lack of transparency in Philippine government administrative processes has far-reaching ethical, social, and professional implications, affecting citizens, public officials, oversight institutions, and IT professionals alike. Fragmented records, inefficient workflows, and limited access to verifiable information not only undermine public trust but also increase opportunities for corruption and weaken democratic governance. Blockchain technology emerges as a viable solution, offering immutable, traceable, and publicly verifiable records that enhance accountability, reduce corruption risk, and improve administrative efficiency.

Implementing a blockchain-based system aligns with global best practices, local legal frameworks such as the Freedom of Information Order and Anti-Red Tape Act, and international development goals including SDG 9. From an ethical perspective, it supports utilitarian, rights-based, and deontological principles by maximizing public good, protecting citizens' rights to accurate information, and reinforcing the government's moral duty for transparency. Professionally, it challenges IT experts to design inclusive, secure, and reliable systems while fostering digital literacy and responsible governance practices.

Overall, the proposed blockchain-enabled solution represents a strategic, ethical, and technically sound approach to strengthening transparency, promoting public accountability, and restoring citizen trust. By combining innovation, legal compliance, and professional integrity, it offers a sustainable pathway toward a more transparent, efficient, and citizen-centered Philippine government.

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